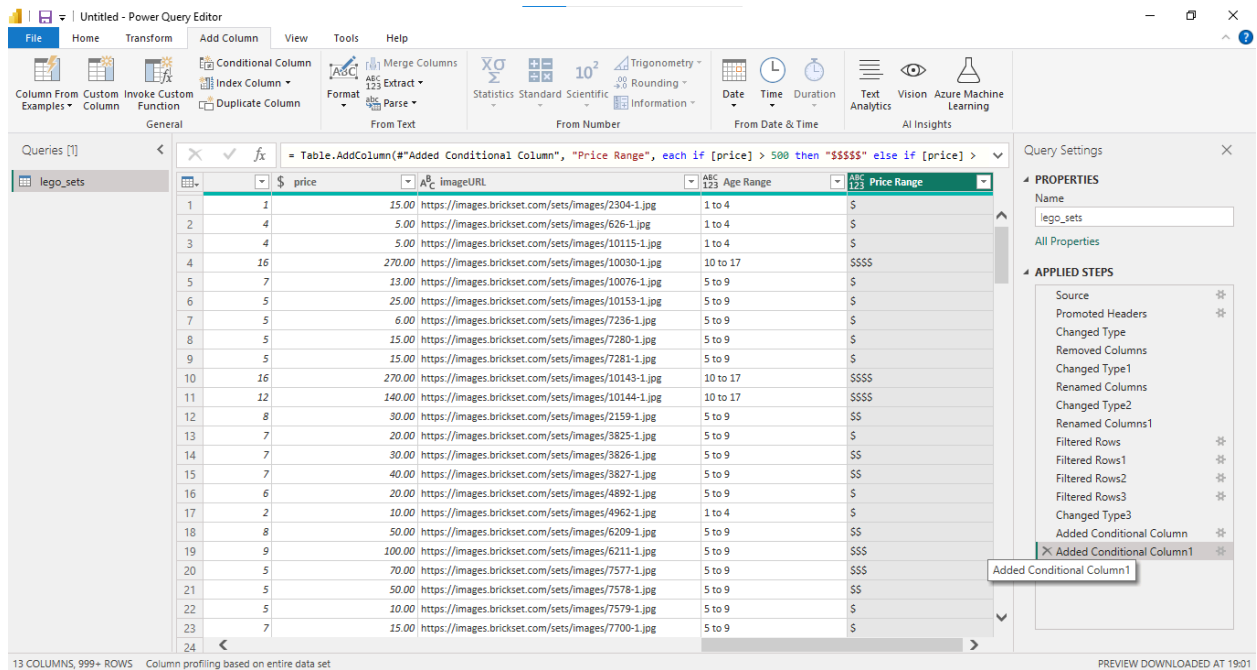


LEGO Sets Project

TASK 1:



The screenshot shows the Power Query Editor interface. The main table has columns: \$ price, imageURL, Age Range, and Price Range. The 'Price Range' column is highlighted, showing values like '\$' and '\$\$\$\$' based on price thresholds. The 'Age Range' column shows values like '1 to 4', '5 to 9', '10 to 17', and '1 to 4'. The 'Price Range' column is highlighted, showing values like '\$' and '\$\$\$\$' based on price thresholds. The 'Age Range' column shows values like '1 to 4', '5 to 9', '10 to 17', and '1 to 4'. The 'Price Range' column is highlighted, showing values like '\$' and '\$\$\$\$' based on price thresholds.

	\$ price	imageURL	Age Range	Price Range
1	15.00	https://images.brickset.com/sets/images/2304-1.jpg	1 to 4	\$
2	5.00	https://images.brickset.com/sets/images/626-1.jpg	1 to 4	\$
3	5.00	https://images.brickset.com/sets/images/10115-1.jpg	1 to 4	\$
4	270.00	https://images.brickset.com/sets/images/10030-1.jpg	10 to 17	\$\$\$\$
5	13.00	https://images.brickset.com/sets/images/10076-1.jpg	5 to 9	\$
6	25.00	https://images.brickset.com/sets/images/10153-1.jpg	5 to 9	\$
7	6.00	https://images.brickset.com/sets/images/7236-1.jpg	5 to 9	\$
8	15.00	https://images.brickset.com/sets/images/7280-1.jpg	5 to 9	\$
9	15.00	https://images.brickset.com/sets/images/7281-1.jpg	5 to 9	\$
10	270.00	https://images.brickset.com/sets/images/10143-1.jpg	10 to 17	\$\$\$\$
11	140.00	https://images.brickset.com/sets/images/10144-1.jpg	10 to 17	\$\$\$\$
12	30.00	https://images.brickset.com/sets/images/2159-1.jpg	5 to 9	\$
13	20.00	https://images.brickset.com/sets/images/3825-1.jpg	5 to 9	\$
14	30.00	https://images.brickset.com/sets/images/3826-1.jpg	5 to 9	\$
15	40.00	https://images.brickset.com/sets/images/3827-1.jpg	5 to 9	\$
16	20.00	https://images.brickset.com/sets/images/4892-1.jpg	5 to 9	\$
17	10.00	https://images.brickset.com/sets/images/4962-1.jpg	1 to 4	\$
18	50.00	https://images.brickset.com/sets/images/6209-1.jpg	5 to 9	\$
19	100.00	https://images.brickset.com/sets/images/6211-1.jpg	5 to 9	\$
20	70.00	https://images.brickset.com/sets/images/7577-1.jpg	5 to 9	\$
21	50.00	https://images.brickset.com/sets/images/7578-1.jpg	5 to 9	\$
22	10.00	https://images.brickset.com/sets/images/7579-1.jpg	5 to 9	\$
23	15.00	https://images.brickset.com/sets/images/7700-1.jpg	5 to 9	\$
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Step 1: Load and prepare the data.

Open Power BI and go to Get Data > select lego(csv) file and load.

Click Home > Transform Data (this opens Power Query Editor)

Remove unnecessary columns and review data types.

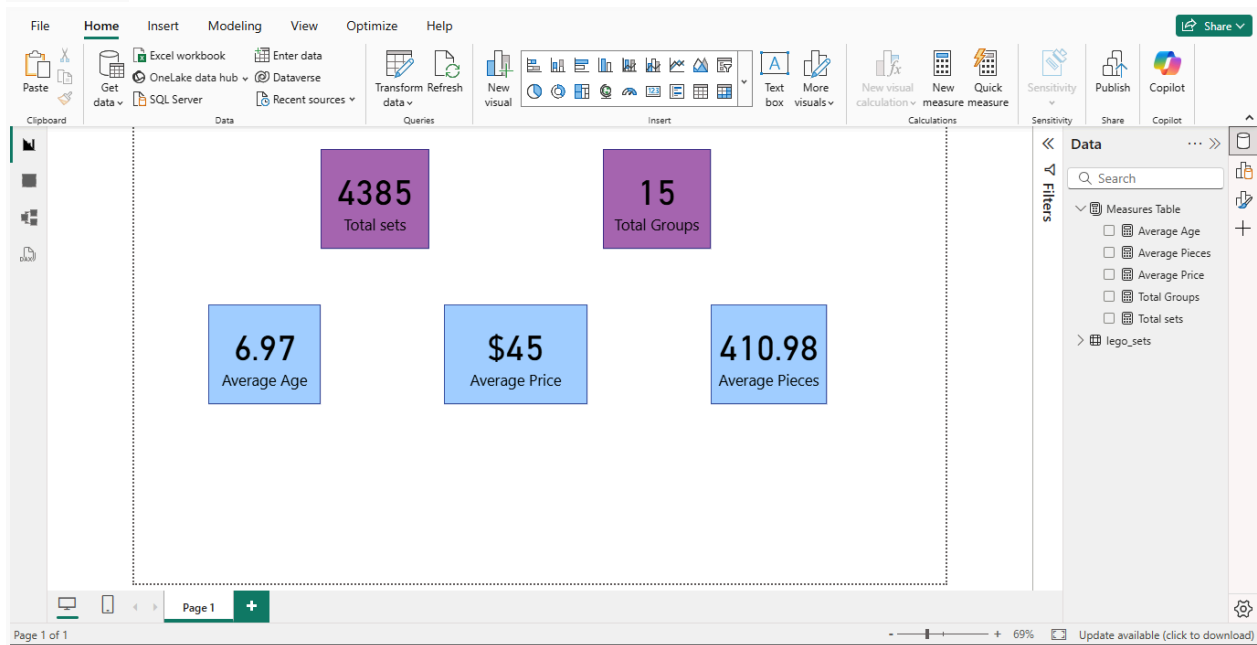
Filter out records with no price, age, pieces or image URL values (Select remove empty from the column header drop down of each column).

Create Conditional Columns to display 'Age Range' and 'Price Range' -

Age Range ("Over 18", "10 to 17", "5 to 9", "1 to 4") and

Price Range (>\$500 = "\$\$\$\$\$", >\$100 = "\$\$\$\$", >\$50 = "\$\$\$", >\$25 = "\$\$", otherwise "\$").

TASK 2:

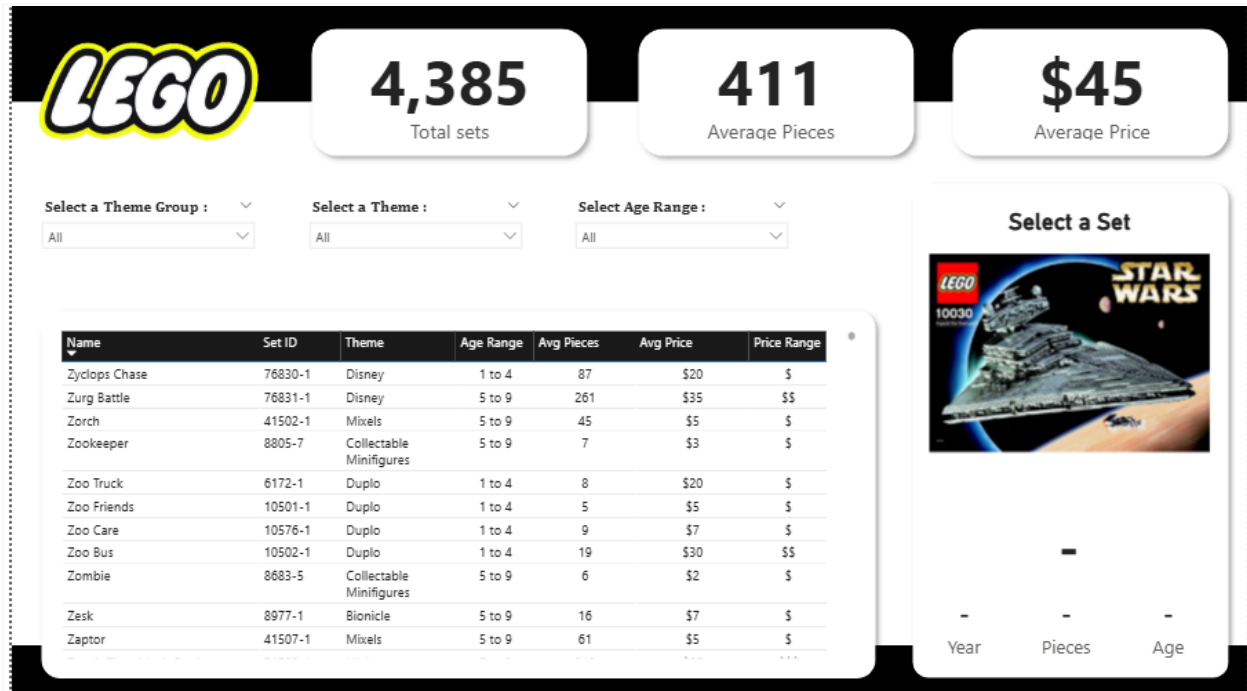


Step 1: Create New Measures (DAX Formulas).

Add measures to calculate -

1. Total sets = `DISTINCTCOUNT(lego_sets[set_id])`
2. Total Groups = `DISTINCTCOUNT(lego_sets[themeGroup])`
3. Average Price = `AVERAGE(lego_sets[price])`
4. Average Age = `AVERAGE(lego_sets[age])`
5. Average Pieces = `AVERAGE(lego_sets[pieces])`

TASK 3:



Step 1 : Insert card visuals

Add KPI cards that show Total Sets, Average Pieces and Average Price.

Step 2 : Add slicers

Add slicers to filter based on the Theme Group, Theme and Age Range.

Step 3 : Insert a table to show filtered data.

Create a table visual and drop the following columns to add data- name, set_id, theme, Age Range, Avg. Pieces, Avg. Price and Price Range.

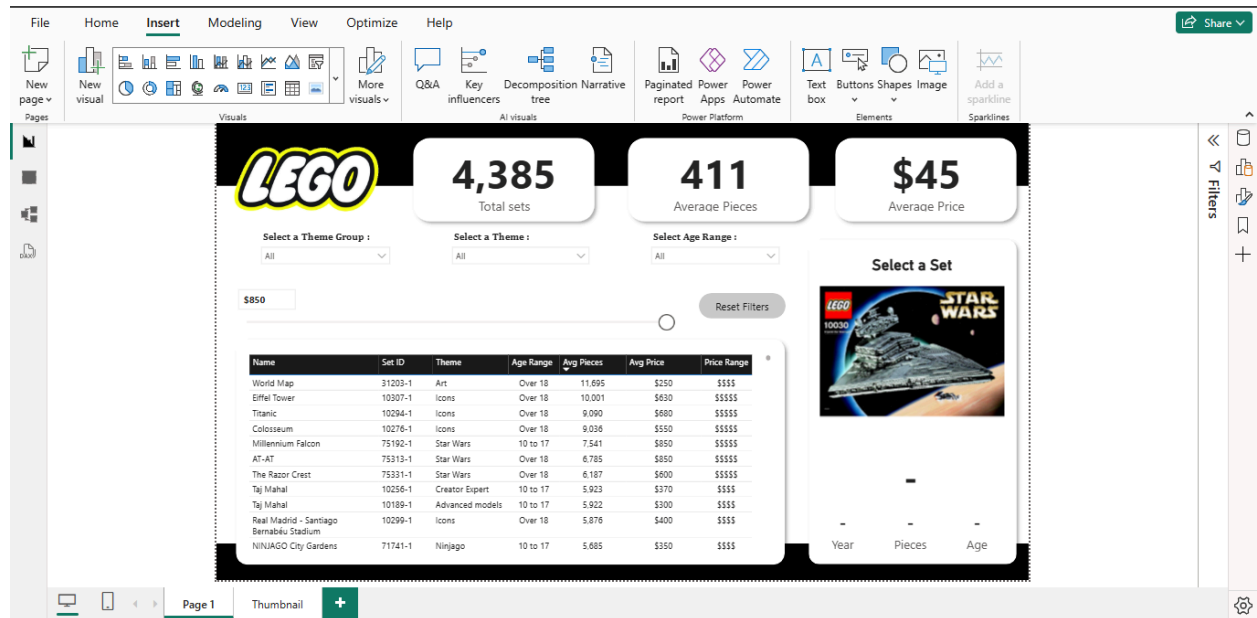
Step 4 : Add a report section to show details for a selected set

Select the 'Simple Image' visual and drag the 'Image url' column. Show details for a selected set including the name, image, price, year, pieces and age using card visuals.

Step 5 : Edit Interactions.

Select the table and go to Format > Edit Interactions. Edit visual interactions to prevent table selections from filtering the top-level cards.

TASK 4



Step 1 : Add a Numeric Range Parameter

Add a Numeric Range Parameter (Max Price) ranging from 0-850 and incrementing by 5.

Add a Slicer to the page and change the style to 'Single Value'

Add a measure comparing average price against the selected slicer value, and use it to filter the table.

Step 2 : Enable tooltips on the table to display the image of each LEGO set on hover

Add a new Report Page named 'Thumbnail'.

Change the page type from 'Standard' to 'Tooltip'.

Insert a new 'simple image' visualisation and drag the image url column to the add data.

Go back to the main report page and select the table visualisation.

Click on Properties > Turn on 'Tooltip' > Select 'Report Page' under type > Select 'Thumbnail' under page box.

Now, you can hover over the lego set details from the table and a thumbnail of the lego set will appear.

Step 3 : Use bookmarks and button actions to allow users to reset all filters on the page

Go to the Insert Tab > Click on Buttons > Select a Blank Button.

Resize and label the text as 'Reset Filters'.

Customize the default and on hover settings of the button by clicking on button style.

Now add a bookmark of the page named 'Unfiltered' (Make sure all the filters are deselected)

Click on the button > Toggle On Action > Select Bookmark under Type > Select 'Unfiltered' from the bookmark drop down.

